1 The strategy

We aim to evaluate a string amplitude (free energy) of the form

$$\mathcal{F} = \int_{\mathcal{M}} d\mu, \tag{1.1}$$

where $\mathcal{M}$ is a suitable moduli space (say of the genus g Riemann surface) and $d\mu$ is a positive measure. The idea is to compare it with

$$\mathcal{F}_0 = \int_{\mathcal{M}} d\mu_0 \tag{1.2}$$

of a solvable model, and express

$$\mathcal{F} = \mathcal{F}_0 \left\langle \frac{d\mu}{d\mu_0} \right\rangle_0, \tag{1.3}$$

where

$$\langle f \rangle_0 = \frac{\int_{\mathcal{M}} d\mu_0 \, f}{\int_{\mathcal{M}} d\mu_0} \tag{1.4}$$

can be numerically evaluated by Monte Carlo sampling of f over $\mathcal{M}$ with probability measure $d\mu_0$.

2 The moduli space

The moduli space $\mathcal{M}_g$ of genus g Riemann surface can be viewed as the base of the obvious fibration

$$\pi : \mathcal{M}_{g,1} \to \mathcal{M}_g, \tag{2.1}$$

and $\mathcal{M}_{g,1}$ can be parameterized via the moduli space $\mathcal{M}_{g,1}^{\text{comb}}$ of metric ribbon graphs, via Kontsevich's isomorphism

$$\mathcal{M}_{g,1} \times \mathbb{R}_+ \xrightarrow{\sim} \mathcal{M}_{g,1}^{\text{comb}}. \tag{2.2}$$

The explicit ribbon graph is constructed by finding the Strebel differential $\omega = \varphi(z) dz^2$, whose zeros of degree d corresponding to vertices of the graph with valence $d+2$. The defining property of ω is such that all of its horizontal trajectories, along which $\varphi(z(s))(\frac{dz}{ds})^2 \in \mathbb{R}_+$, are closed or end on a zero. The length of a closed horizontal trajectory γ is equal to

$$\ell(\gamma) = \oint_\gamma \sqrt{\varphi(z)} \, dz, \tag{2.3}$$

and is constant along a continuous family of closed horizontal trajectories. In the case of one puncture, all closed horizontal trajectories are homologous and enclose the puncture where $\varphi(z)$ has a double pole.

At a generic point in the moduli space, the ribbon graph in question has only cubic vertices, and Euler characteristic $V - E + F = 2 - 2g$ with $3V = 2E$ and $F = 1$. So the number of vertices is $V = 4g - 2$, and the number of edges is $E = 6g - 3$.

Consider the example of a 1-punctured torus, parameterized by the holomorphic coordinate $z \sim z + 2\pi \sim z + 2\pi\tau$ with the puncture at $z = 0$. The Strebel differential is of the form

$$\omega = \ell^2 \left(\mathcal{P}_\tau(z) - a\right) dz^2, \tag{2.4}$$

where $\mathcal{P}_\tau(z)$ is the Weierstrass elliptic function which has a double zero at $z = 0$,

$$\mathcal{P}_\tau(z) = \frac{1}{z^2} + \sum_{(m,n)\neq(0,0)} \left[\frac{1}{(z - 2\pi(m + n\tau))^2} - \frac{1}{(2\pi(m + n\tau))^2}\right]. \tag{2.5}$$

The constant a is determined by τ by the condition that the horizontal trajectories are closed.

Rather than mapping metric graph data to plumbing parameters and computing CFT correlators from conformal blocks, the strategy is to directly computing CFT observable in terms of the metric graph data. For instance, the torus partition function of a free boson X can be represented, modulo Weyl anomaly, as

$$\begin{aligned} Z(\theta_1, \theta_2) = \int [DX(\sigma)]_{0<\sigma<2\pi} \Psi_1[X(\sigma)] \prod_{0<\sigma<\theta_1} \delta(X(\sigma) - X(\pi + \theta_1 - \sigma)) \\ \times \prod_{\theta_1<\sigma<\theta_2} \delta(X(\sigma) - X(\pi + \theta_1 + \theta_2 - \sigma)) \prod_{\theta_2<\sigma<\pi} \delta(X(\sigma) - X(2\pi + \theta_2 - \sigma)). \end{aligned} \tag{2.6}$$

Here $0 < \theta_1 < \theta_2 < 2\pi$ are related to the lengths ℓ_1, ℓ_2, ℓ_3 of the metric graph by $\theta_1 = \pi \frac{\ell_1}{\ell_1+\ell_2+\ell_3}$ and $\theta_2 = \pi \frac{\ell_1+\ell_2}{\ell_1+\ell_2+\ell_3}$. $\Psi_1[X(\sigma)]$ is the wave functional of the identity operator. As in (2.8.23) of Polchinski,

$$\Psi_1[X(\sigma)] \propto \exp\left(-\frac{1}{\alpha'} \sum_{m\geq 1} m X_m X_{-m}\right), \tag{2.7}$$

where X_m is the m-th Fourier mode of $X(\sigma)$.

To obtain the genus 2 partition function, the ribbon graph at a generic point in the moduli space has 6 cubic vertices and 9 edges. The partition function can be obtained by integrating $\Psi_1[X(\sigma)]$ with the identification of X on pairs of 18 segments of the unit circle.

3 Application to $c = 1$ string theory

The idea is to take $d\mu_R$ to be the measure associated with the free energy of $c = 1$ string theory at inverse temperature $\beta = 2\pi R$, and compare the measure at different values of R. Explicitly, as in (4.97) of the string notes, we have on $\mathcal{M}_g$

$$d\mu_R = \prod_{k=1}^{3g-3} i d\tau^k \wedge d\bar{\tau}^k \left\langle \prod_{k=1}^{3g-3} \mathcal{B}_{\bar{\tau}^k} \mathcal{B}_{\tau^k} \right\rangle_\Sigma , \tag{3.1}$$

which is a positive measure. The correlator is evaluated for the matter CFT that consists of $c = 25$ Liouville theory and a compact boson of radius R. It follows that

$$\frac{d\mu_R}{d\mu_{R_0}} = \frac{Z_R(\Sigma)}{Z_{R_0}(\Sigma)}, \tag{3.2}$$

where $Z_R(\Sigma)$ is the partition function of the compact boson of radius R on the Riemann surface Σ. Note that the Weyl anomaly cancels between the numerator and the denominator and so the RHS is well-defined.

Typically, the RHS of (3.2) is not hard to compute. The main challenge is to find an algorithm for sampling $\mathcal{M}_g$, say with Metropolis algorithm, subject to the probability measure $d\mu_{R_0}$.

In the case of the genus 2 free energy, there is an expression in terms of the period matrix Ω_{IJ},

$$\mathcal{F}_2(R) = \int_{\mathcal{M}_2} |\Phi(\Omega) d\Omega_{11} d\Omega_{22} d\Omega_{12}|^2 \frac{V_{26}}{(4\pi^2\alpha')^{26} (\det \mathrm{Im}\Omega)^{13} Z_{26}(\Sigma)} Z_{1,R}(\Sigma) Z_L(\Sigma), \tag{3.3}$$

where $\Phi(\Omega) = 2\pi i/\chi_{10}(\Omega)$ as in (4.109) of the string notes, $Z_{26}(\Sigma)$ is the genus two partition function of 26 noncompact free bosons with target space volume V_{26}, $Z_{1,R}(\Sigma)$ is the partition function of a compact boson at radius R, and $Z_L(\Sigma)$ is the $c = 25$ Liouville CFT partition function. Each partition function is subject to the Weyl anomaly, which cancels in the combination $Z_{1,R}(\Sigma) Z_L(\Sigma)/Z_{26}(\Sigma)$.

4 Ψ_1 of Liouville CFT

The Euclidean action of Liouville theory is

$$S_L = \frac{1}{4\pi} \int d^2\sigma \sqrt{g} (g^{ab} \partial_a \partial_b \phi + QR\phi + 4\pi\mu e^{2b\phi}), \tag{4.1}$$

where $Q = b + b^{-1}$. On the cylinder, the Hamiltonian reads

$$H = \int d\sigma \left[\pi \Pi_\phi^2 + \frac{1}{4\pi} (\partial_\sigma \phi)^2 + : \mu e^{2b\phi} : \right], \tag{4.2}$$

where : : is the normal ordering defined via the mode expansion

$$\begin{aligned} \phi &= \sum_{n \in \mathbb{Z}} \frac{1}{\sqrt{2|n|}} (a_n + a_{-n}^\dagger) e^{in\sigma}, \\ \Pi_\phi &= \frac{-i}{2\pi} \sum_{n \in \mathbb{Z}} \sqrt{\frac{|n|}{2}} (a_n - a_{-n}^\dagger) e^{in\sigma}. \end{aligned} \tag{4.3}$$

The exactness of this Hamiltonian can be justified through its commutation relation with the Virasoro generators defined by the holomorphic stress-energy tensor of the form

$$T = - : \partial\phi\partial\phi : + Q\partial^2\phi. \tag{4.4}$$

We could consider alternatively the regularized Hamiltonian defined by discretizing space,

$$H = \sum_{i=1}^{N} \left[\frac{N}{2} \Pi(i)^2 + \frac{N}{2(2\pi)^2} (\phi(i+1) - \phi(i))^2 + \mu \frac{2\pi}{N} e^{2b\phi(i)} \right], \tag{4.5}$$

where $\phi(N+1) \equiv \phi(1)$, $[\phi(i), \Pi(j)] = i\delta_{ij}$, and seek the identity wave function $\Psi_1(\phi(1), \cdots, \phi(N))$.

Note however that Ψ_1 is not the ground state, but rather the analytic continuation of a scattering state of real Liouville momentum P to $P = i\frac{Q}{2}$. The energy of Ψ_1 on the cylinder is $H = -\frac{1}{12} - \frac{Q^2}{2}$. Moreover, Ψ_1 is not normalizable and blows up exponentially in the large field region $\phi \to -\infty$ as $\Psi_1 \sim e^{-\frac{Q}{2}\phi}$. This leads to a divergence in the zero mode integration that is likely cancelled by the Weyl anomaly due to the Weyl frame of the metric associated with the ribbon graph/Strebel differential.

5 The partition function of free boson CFT from Ψ_1

Discretizing the unit circle into L sites, we consider the approximation of (2.7),

$$\begin{aligned} \Psi_1[X] &\propto \exp\left[-\frac{1}{\pi\alpha' L} \sum_{m=1}^{L} \sin\frac{\pi m}{L} \sum_{k,\ell=1}^{L} X(k) X(\ell) e^{2\pi i \frac{m}{L}(k-\ell)} \right] \\ &= \exp\left[-\frac{1}{\pi\alpha' L} \sum_{k,n=1}^{L} X(n) X(n+k) \frac{\sin\frac{\pi}{L}}{\cos\frac{2\pi k}{L} - \cos\frac{\pi}{L}} \right]. \end{aligned} \tag{5.1}$$

We will assume that L is even. Next we split $\frac{L}{2} = \ell_1 + \ell_2 + \ell_3$, where each ℓ_i is a positive integer, and make the identification

$$\begin{aligned} &X(\tfrac{L}{2} + \ell_1 + 1 - n) = X(n), \quad 1 \le n \le \ell_1, \\ &X(\tfrac{L}{2} + 2\ell_1 + \ell_2 + 1 - n) = X(n), \quad \ell_1 + 1 \le n \le \ell_1 + \ell_2, \\ &X(L + \ell_1 + \ell_2 + 1 - n) = X(n), \quad \ell_1 + \ell_2 + 1 \le n \le \tfrac{L}{2}, \end{aligned} \tag{5.2}$$

as a discretized version of (2.6). Finally, we perform the integration with respect to the remaining independent variables $X(n)$, $1 \le n \le \frac{L}{2}$.

After imposing the constraint (5.2), we can write $\Psi_1[X]$ as proportional to

$$\exp\left[-\sum_{k,\ell=1}^{\frac{L}{2}} A_{k\ell} X(k) X(\ell)\right], \tag{5.3}$$

where the eigenvalues of the matrix $A_{k\ell}$ are positive except for a single zero. The integration over $X(k)$, $1 \le k \le \frac{L}{2}$ then produces, up to a constant normalization factor, the determinant

$$Z_L(t_1, t_2, t_3) = (\det{}' A_{k\ell})^{-\frac{1}{2}}, \tag{5.4}$$

where $t_i \equiv 2\ell_i/L$ with $t_1 + t_2 + t_3 = 1$, and the prime stands for removing the zero mode. This is explicitly evaluated in "discretized wave functional.nb" in two different ways that give the same answer.

In the limit $L \to \infty$, Z_L is divergent. There are two sources for the divergence. One is due to the discretization scheme and the normalization of the Gaussian measure, and should go like $\exp(const \cdot L)$. The other is due to the Weyl anomaly associated with the surface defined by the Strebel differential, and is universal. In particular, namely a cubic vertex of the ribbon graph, the metric goes like $ds^2 = e^{2\omega} dz d\bar{z} \sim |z| dz d\bar{z}$, and so $\omega \sim \frac{1}{2}\log|z|$. Need a puncture, on the otherhand, we have $\omega \sim -\log|z|$. The Weyl anomaly factor in going from the flat metric to the ribbon graph metric on the torus is e^{S_L} with the linear dilaton action S_L given simply by

$$S_L = \frac{1}{12\pi} \int_{T^2} d^2 z\, \partial\omega \bar{\partial}\omega. \tag{5.5}$$

As $\partial\omega \sim \frac{1}{4z}$ near a cubic vertex, and $\partial\omega \sim -\frac{1}{2z}$ near a puncture, we get logarithmic divergence

$$S_L \sim \frac{1}{12\pi} \cdot 2 \cdot 2\pi \left(2 \cdot \frac{1}{4^2} \log\frac{1}{\delta} + \frac{1}{2^2} \log\frac{1}{\epsilon}\right), \tag{5.6}$$

where δ and ϵ are the cutoff parameters near the pair of cubic vertex points and the puncture respectively. Note that the coefficient of the log divergence is independent from the moduli.

5.1 Torus one point function

Let's consider the torus one point function, for example of $\partial X \bar{\partial} X$. This is a weight $(1,1)$ primary operator, so

$$\langle : \partial X \bar{\partial} X : (0) \rangle_{T^2(t_1,t_2)} = |\widehat{f}(0)|^{-2} \langle : \partial X \bar{\partial} X : (0) \rangle_{D^2(t_1,t_2)}, \tag{5.7}$$

where the correlator on the RHS can be approximately computed as

$$\propto \int \left[\prod_{k=1}^{L} dX(k) \right] \Psi_{\partial X \bar{\partial} X}[X(k)] \delta_{\text{identification}}, \tag{5.8}$$

with $\delta_{\text{identification}}$ corresponding to the rules (5.2). The wavefunctional corresponding to the (normal ordered) operator $\partial X \bar{\partial} X$ inserted at the origin of the disk is given by Polchinski (2.8.28) and the paragraph below it as

$$\Psi_{\partial X \bar{\partial} X}[X(\sigma)] \propto X_1 X_{-1} \exp\left(-\frac{1}{\alpha'} \sum_{m \geq 1} m X_m X_{-m} \right). \tag{5.9}$$

The gaussian factor is the same as in (2.7), while the prefactor can be approximated in the discretization scheme as

$$X_1 X_{-1} \sim \frac{1}{L^2} \sum_{k,\ell=1}^{L} X(k) X(\ell) e^{\frac{2\pi i}{L}(k-\ell)}, \tag{5.10}$$

which we can denote after applying the identification rules as

$$X_1 X_{-1} \delta_{\text{identification}} \sim \sum_{k,\ell=1}^{L/2} C_{k\ell} X(k) X(\ell). \tag{5.11}$$

If we denote the shift used in the identification rule (5.2) as

$$s(n) = \begin{cases} \frac{L}{2} + \ell_1 + 1, & 1 \leq n \leq \ell_1 \\ \frac{L}{2} + \ell_2 + 2\ell_1 + 1, & \ell_1 + 1 \leq n \leq \ell_1 + \ell_2 \\ \frac{L}{2} + \ell_3 + 2\ell_2 + 2\ell_1 + 1, & \ell_1 + \ell_2 + 1 \leq n \leq \frac{L}{2} \end{cases} \tag{5.12}$$

then

$$C_{k\ell} = \frac{1}{L^2} \left(e^{\frac{2\pi i}{L}(k-\ell)} + e^{-\frac{2\pi i}{L}(k-\ell-s(k)+s(\ell))} + e^{\frac{2\pi i}{L}(k+\ell-s(\ell))} + e^{-\frac{2\pi i}{L}(k+\ell-s(k))} \right). \tag{5.13}$$

Finally, we wish to evaluate

$$\begin{aligned}&\int\left[\prod_{k=1}^{L/2}dX(k)\right]\exp\left(-\sum_{k,\ell=1}^{L/2}A_{k\ell}X(k)X(\ell)\right)\sum_{r,s=1}^{L/2}C_{rs}X(r)X(s)\\&=(\det{}'A)^{-\frac{1}{2}}\frac{1}{2}\mathrm{tr}\,'\left(CA^{-1}\right).\end{aligned}\tag{5.14}$$

Recall that A has a zero eigenvector, namely $\sum_{k=1}^{L/2}X(k)$, so A^{-1} is undefined. However recall that $\sum_{k,\ell}X(k)X(\ell)A_{k\ell}$ is independent of this zero mode, and one can see that $\sum_{k=1}^{L/2}C_{k\ell}=0=\sum_{k=1}^{L/2}C_{\ell k}$ for any ℓ, so that $\sum_{k,\ell}X(k)X(\ell)C_{k\ell}$ is independent of this zero mode as well. Therefore we can consider the "reduced" $(\frac{L}{2}-1)\times(\frac{L}{2}-1)$ matrices $\widetilde{A}$ and $\widetilde{C}$ defined by imposing the additional constraint

$$X(\frac{L}{2})=-\sum_{k=1}^{L/2}X(k)\tag{5.15}$$

after which $\widetilde{A}$ is invertible and

$$\begin{aligned}&\int\left[\prod_{k=1}^{L/2}dX(k)\right]\exp\left(-\sum_{k,\ell=1}^{L/2}A_{k\ell}X(k)X(\ell)\right)\sum_{r,s=1}^{L/2}C_{rs}X(r)X(s)\\&=(\text{const.})\int\left[\prod_{k=1}^{L/2-1}dX(k)\right]\exp\left(-\sum_{k,\ell=1}^{L/2-1}\widetilde{A}_{k\ell}X(k)X(\ell)\right)\sum_{r,s=1}^{L/2-1}\widetilde{C}_{rs}X(r)X(s)\\&=\sqrt{\frac{L}{2}}\,(\det\widetilde{A})^{-\frac{1}{2}}\frac{1}{2}\mathrm{tr}\,\left(\widetilde{C}\widetilde{A}^{-1}\right),\end{aligned}\tag{5.16}$$

with the proportionality factor $\frac{L}{2}\det{}'A=\det\widetilde{A}$ determined experimentally.

We can now consider

$$(Z(\tau(t_1,t_2)))^{-1}\langle:\partial X\bar{\partial}X:(0)\rangle_{T^2(t_1,t_2)}=|\widehat{f}(0)|^{-2}\frac{\langle:\partial X\bar{\partial}X:(0)\rangle_{D^2(t_1,t_2)}}{\langle 1\rangle_{D^2(t_1,t_2)}}\sim\frac{1}{2}|\widehat{f}(0)|^{-2}\mathrm{tr}\,(\widetilde{C}\widetilde{A}^{-1}),\tag{5.17}$$

where the determinants, and any Weyl anomaly factors should cancel between the numerator and denominator of the RHS calculated in the discretization scheme. Finally, from Polchinski (7.2.16) but with $\partial_{w'}\to\partial_{\bar{w}'}$, we find that **[am I missing terms here from e.g. $\partial_w\vartheta_1(\bar{w}|\tau)|_{w\to 0}$?]**

$$(Z(\tau))^{-1}\langle:\partial X\bar{\partial}X:(0)\rangle_{T^2(\tau)}=-\frac{\alpha'}{8\pi\tau_2},\tag{5.18}$$

and get that

$$\mathrm{tr}\,(\widetilde{C}\widetilde{A}^{-1})\propto\frac{|\widehat{f}(0)|^2}{\tau_2},\tag{5.19}$$

which is modular invariant.

[Reconcile with the above:]
We find that, to good numerical accuracy,

$$|\widehat{f}(0)|^{-2}\left(\frac{1}{2\pi}-\mathrm{tr}\,(\widetilde{C}\widetilde{A}^{-1})\right)=\frac{1}{2\tau_2} \tag{5.20}$$

5.2 Compact Boson

Consider the torus partition function of a compact boson X with radius $X \sim X + 2\pi R$. On a unit circle, a configuration of X can be parametrized by the constant, the winding mode, and nonzero Fourier modes X_m:

$$X_b(\sigma) = x_0 + p\sigma R + \sum_{m=-\infty}^{\infty} X_m e^{im\sigma} \tag{5.21}$$

where $p \in \mathbb{Z}$ is the winding number. For any configuration with nonzero winding number, its classical action will be logarithmically divergent, and the wavefunctional $\Psi_1[X_b] = \int [\mathcal{D}X]_{X_b} e^{-S[X]}$ will be 0. Therefore, the wavefunctional $\Psi_1[X_b]$ is identical to that of a non-compact boson:

$$\Psi_1[p, X_m] \propto \delta_{p,0} \exp\left[-\frac{1}{\alpha'}\sum_{m\geq 1} m X_m X_{-m}\right] \tag{5.22}$$

which can be discretized as:

$$\begin{aligned}\Psi_1[X] &= \exp\left(-\frac{1}{\pi\alpha' L}\sum_{k,n=1}^{L} X(n)X(n+k)\frac{\sin\frac{\pi}{L}}{\cos\frac{2\pi k}{L}-\cos\frac{\pi}{L}}\right)\\ &\equiv \exp\left(-\frac{1}{\pi}\sum_{k,l=1}^{L} M_{kl}X(k)X(l)\right)\end{aligned} \tag{5.23}$$

The torus partition function can be computed as:

$$Z(l_1, l_2, l_3; R) = \sum_{s_1,s_2,s_3}\int [\mathcal{D}X]\Psi_1[X]\delta_{s_1,s_2,s_3}(\ldots) \tag{5.24}$$

where the delta function imposes the following identification:

$$\begin{aligned}&X(L/2+l_1+1-n) = X(n) + 2\pi R s_1, \quad 1 \leq n \leq l_1,\\ &X(L/2+2l_1+l_2+1-n) = X(n) + 2\pi R s_2, \quad l_1+1 \leq n \leq l_1+l_2,\\ &X(L+l_1+l_2+1-n) = X(n) + 2\pi R s_3, \quad l_1+l_2+1 \leq n \leq L/2\end{aligned} \tag{5.25}$$

This can be discretized as:

$$Z(l_i;R) = \int \prod_{i=1}^{L/2} dX(i) \exp\left[-\frac{1}{\pi}\left(\sum_{k,l=1}^{L/2} A_{kl}X(k)X(l) - 2\sum_{k=1}^{L/2} v_k X(k) - C\right)\right] \tag{5.26}$$

where:

$$\begin{aligned} A_{kl} &= M_{kl} + M_{k'l} + M_{kl'} + M_{k'l'} \\ v_k &= 2\pi R \sum_{i=1}^{3} s_i \sum_{m\in \text{seg } i} (M_{km'} + M_{k'm'}) \\ C &= 4\pi^2 R^2 \sum_{i,j}^{3} s_i s_j \sum_{k\in \text{seg } i} \sum_{m\in \text{seg } j} M_{k'm'} \end{aligned} \tag{5.27}$$

and the primed position k' stands for the position on the other half circle that's identified with k.

Now we explicitly perform the Gaussian integral. The matrix A has a zero mode given by $\sum_{k=1}^{L/2} X(k)$. Since

$$\begin{aligned} \sum_{k=1}^{L/2} v_k &= 2\pi R \sum_i s_i \sum_{k=1}^{L/2} \sum_{m\in \text{seg } i} (M_{km'} + M_{k'm'}) \\ &= 2\pi R \sum_i s_i \sum_{m\in \text{seg } i} \sum_{k=1}^{L} M_{km'} = 0 \end{aligned} \tag{5.28}$$

the shift v_k is then orthogonal to the zero mode, and hence the zero mode can be factored out, giving a factor of $2\pi R$. In the subspace with nonzero eigenvalue of A, we define the reduced $(L/2-1)\times(L/2-1)$ matrix A' and the reduced $(L/2-1)$-dimensional vector v' by imposing the constraint $X(L/2) = -\sum_{k=1}^{L/2-1} X(k)$:

$$\begin{aligned} A'_{kl} &= A_{kl} - A_{k,\frac{L}{2}} - A_{\frac{L}{2},l} + A_{\frac{L}{2},\frac{L}{2}} \\ v'_k &= v_k - v_{\frac{L}{2}} \end{aligned} \tag{5.29}$$

After performing the Gaussian integral in this subspace, the partition function is given by:

$$\begin{aligned} Z(l_i;R) &= 2\pi R(\det A')^{-1/2} \sum_{s_1,s_2,s_3} \exp(-\pi^{-1}C + \pi^{-1} \sum_{k,l=1}^{L/2-1} v'_k (A')^{-1}_{kl} v'_l) \\ &= 2\pi R(\det A')^{-1/2} \sum_{s_1,s_2,s_3} \exp(-4\pi R^2 \sum_{i,j} s_i s_j T_{ij}) \\ &= 2\pi R(\det A')^{-1/2} \Theta(4iR^2 T) \end{aligned} \tag{5.30}$$

where $\Theta(T)$ is the Siegel theta function of matrix T. This can be evaluated using the builtin Mathematica function `SiegelTheta[T, 0]` to evade explicit evaluation of infinite sums. The matrix T is defined as:

$$\begin{aligned} T_{ij} &= \sum_{m\in \text{seg } i} \sum_{n\in \text{seg } j} \left(M_{m'n'} - \sum_{k,l=1}^{\frac{L}{2}-1} (A')^{-1}_{st} W_{km} W_{ln} \right) \\ W_{km} &= M_{km'} + M_{k'm'} - M_{\frac{L}{2},m'} - M_{(\frac{L}{2})',m'} \end{aligned} \tag{5.31}$$

Since the $Z(l_i, R)$ only depend on R via the zero mode factor $2\pi R$ and the theta function, in order to check the T-duality relation $Z(l_i; R) = Z(l_i; R^{-1})$ (taking $\alpha' = 1$), we only need to check:

$$R\Theta(4iR^2T) = R^{-1}\Theta(4iR^{-2}T) \tag{5.32}$$

6 Moduli and Weyl anomaly in disc coordinate

Let $z = e^{-iw}$ with the unit disc parameterized by $|z| < 1$ or $\text{Im}(w) > 0$. The holomorphic 1-form on the torus can be expressed in the disc coordinate as

$$f(z)dz = \sum_{n=1}^{\infty} a_n z^{n-1} dz = -i \sum_{n=1}^{\infty} a_n e^{-inw} dw. \tag{6.1}$$

We can normalize with $a_1 = 1$, and solve $a_2, \cdots, a_L$ numerically by imposing

$$\begin{aligned} \sum_{n=1}^{\frac{L}{2}} a_n e^{\frac{2\pi i}{L}(\frac{L}{2}+\ell_1+1-k)n} &= -\sum_{n=1}^{\frac{L}{2}} a_n e^{\frac{2\pi i}{L}kn}, \qquad 1 \le k \le \ell_1, \\ \sum_{n=1}^{\frac{L}{2}} a_n e^{\frac{2\pi i}{L}(\frac{L}{2}+2\ell_1+\ell_2+1-k)n} &= -\sum_{n=1}^{\frac{L}{2}} a_n e^{\frac{2\pi i}{L}kn}, \qquad \ell_1+1 \le k \le \ell_1+\ell_2, \\ \sum_{n=1}^{\frac{L}{2}} a_n e^{\frac{2\pi i}{L}(L+\ell_1+\ell_2+1-k)n} &= -\sum_{n=1}^{\frac{L}{2}} a_n e^{\frac{2\pi i}{L}kn}, \qquad \ell_1+\ell_2+1 \le k \le \tfrac{L}{2}, \end{aligned} \tag{6.2}$$

The relative sign is due to flipping the orientation on dw in the identification. Evaluating the periods of $f(z)dz$ along α and β cycle then determines τ in terms of ℓ_1, ℓ_2, ℓ_3.

The flat coordinate on the torus (up to a constant rescaling) can be written as

$$u = \int_{z_0}^{z} f(z')dz' \tag{6.3}$$

The metric associated with the ribbon graph is

$$\frac{dzd\bar{z}}{|z|^2} = \frac{dud\bar{u}}{|z\partial_z u|^2}, \tag{6.4}$$

and so the Weyl factor is given by

$$\omega = -\log|zf(z)|\,. \tag{6.5}$$

(5.5) is evaluated as

$$\begin{aligned} S_L &= \frac{1}{12\pi}\int_D d^2z\partial_z\omega\bar{\partial}_{\bar{z}}\omega \\ &= \frac{1}{12\pi i}\oint_{\partial D} dz\,\omega\partial_z\omega, \end{aligned} \tag{6.6}$$

where D is the unit disc with the small neighborhoods of the singular points cut out. The divergent contribution from the component of ∂D near $z = 0$ is moduli independent (as $f(0) = 1$). Near one of the cubic vertex i.e. zero of Strebel differential, $u = u_i$ $(i = 1, 2)$, we have $\partial_z u = f(z) \approx b_i(u - u_i)^{-\frac{1}{2}}$, and $\partial_z f(z) = f(z)\partial_u f(z) \approx -\frac{1}{2}b_i^2(u - u_i)^{-2}$. It follows that

$$du\,\omega\partial_u\omega \approx -\frac{du}{4(u-u_i)}\log|f(z)| \tag{6.7}$$

near $u = u_i$. Therefore we have

$$S_L = \frac{\log|b_1| + \log|b_2|}{24} + \text{universal divergence.} \tag{6.8}$$

Here the cutoff is assumed to be $|u - u_i| > \delta$. But should we impose the cutoff on z instead? Near $u = u_i$, we have $z - z_i \approx \frac{2}{3b_i}(u - u_i)^{\frac{3}{2}}$, and $f(z) \approx b_i(\frac{3b_i}{2})^{-\frac{1}{3}}(z - z_i)^{-\frac{1}{3}}$. If we impose the cutoff in z, which would seem to be the more compatible with the discretization scheme on the disc, the moduli dependent part of the Weyl anomaly would be given by

$$S_L = \frac{\log|b_1| + \log|b_2|}{36} + \text{universal divergence.} \tag{6.9}$$

7 The bc system

The identity wave functional of bc system on the unit circle can be written as

$$\Psi_1[c(\sigma)] \propto \prod_{n=2}^{\infty} c_n, \tag{7.1}$$

where c_n is the n-th Fourier mode of $c(\sigma)$ (up to a factor of i due to the conformal transformation between cylinder and plane). In the disc coordinate, we can write $c(z) = \sum_{n\in\mathbb{Z}} c_n z^{1-n}$.

It follows that $c(z)\Psi_1$ extends holomorphically to the interior of the unit disc, as expected. The analogous ground state wave functionals are

$$\Psi_c = c_1\Psi_1 \propto \prod_{n=1}^{\infty} c_n, \quad \Psi_{c\partial c} = c_1 c_0 \Psi_1 \propto \prod_{n=0}^{\infty} c_n. \tag{7.2}$$

[Q: is there a way to make BRST symmetry manifest for the discretized matter+ghost system?]

Now discretizing the unit circle into L sites, we have the approximation

$$\Psi_1[c] \propto \prod_{m=2}^{\frac{L}{2}} \sum_{k=1}^{L} c(k) e^{2\pi i \frac{mk}{L}}. \tag{7.3}$$

Next, we identify pairs of $c(k)$'s according to

$$\begin{aligned} &c(\tfrac{L}{2}+\ell_1+1-n) = -c(n), \quad 1 \le n \le \ell_1, \\ &c(\tfrac{L}{2}+2\ell_1+\ell_2+1-n) = -c(n), \quad \ell_1+1 \le n \le \ell_1+\ell_2, \\ &c(L+\ell_1+\ell_2+1-n) = -c(n), \quad \ell_1+\ell_2+1 \le n \le \tfrac{L}{2}, \end{aligned} \tag{7.4}$$

leaving $\frac{L}{2}$ independent variables $c(k)$, $k = 1, \cdots, \frac{L}{2}$. We then integrate the constrained $\Psi_1[c]$ over these $\frac{L}{2}$ independent variables. To obtain a nonzero result requires removing the constant mode of c.

Let us consider the constrained $\Psi_c[c]$, which can be put in the form

$$\prod_{m=1}^{\frac{L}{2}} \sum_{k=1}^{\frac{L}{2}} B_{mk} c(k). \tag{7.5}$$

The $\frac{L}{2} \times \frac{L}{2}$ matrix B is given by

$$\begin{aligned} B_{mn} &= e^{2\pi i \frac{mn}{L}} - e^{2\pi i \frac{m}{L}(\frac{L}{2}+\ell_1+1-n)}, \quad 1 \le n \le \ell_1, \\ B_{mn} &= e^{2\pi i \frac{mn}{L}} - e^{2\pi i \frac{m}{L}(\frac{L}{2}+2\ell_1+\ell_2+1-n)}, \quad \ell_1+1 \le n \le \ell_1+\ell_2, \\ B_{mn} &= e^{2\pi i \frac{mn}{L}} - e^{2\pi i \frac{m}{L}(L+\ell_1+\ell_2+1-n)}, \quad \ell_1+\ell_2+1 \le n \le \tfrac{L}{2}. \end{aligned} \tag{7.6}$$

The result of integrating (7.5) over $c(k)$, $k = 1, \cdots, \frac{L}{2}$, is $\det B$.

The question however is what is the relation between $\det B$ and the torus correlator with appropriate b, c insertions. While there is a c insertion at the center of the disc, there should be a b insertion at the boundary of the disc. Moreover, the b insertion should be localized at

the singular points on the boundary of disc where the different segments join. We are after the moduli integration measure

$$dt_1 dt_2 \langle \mathcal{B}_{t_1} \mathcal{B}_{t_2} c\widetilde{c} \rangle_{T^2(t_1,t_2)} \tag{7.7}$$

Transforming from the flat coordinate to the disc coordinate via (6.3), we end up with

$$dt_1 dt_2 \langle \mathcal{B}_{t_1} \mathcal{B}_{t_2} c\widetilde{c}(0) \rangle_{D^2(t_1,t_2)} |\widehat{f}(0)|^2, \tag{7.8}$$

where $\widehat{f}(z)$ is proportional to $f(z)$ of (6.1), normalized such that the period of $\widehat{f}(z)dz$ along the α-cycle is 1.

If we include bc system with 26 free bosons X^μ, so that the Weyl anomalies cancel, we expect the disc frame correlator to give

$$\langle \mathcal{B}_{t_1} \mathcal{B}_{t_2} c\widetilde{c}(0) \rangle^{X^\mu bc}_{D^2(t_1,t_2)} \propto \frac{d\tau d\bar{\tau}}{dt_1 dt_2} |\widehat{f}(0)|^{-2} \tau_2^{-13} |\eta(\tau)|^{-48}. \tag{7.9}$$

The LHS is calculated in the discretized model, up to a moduli-independent normalization factor (that depends on L), as

$$(\det{}' A)^{-13} |\det B|^2. \tag{7.10}$$

7.1 Dependence on c position

Instead of inserting the c ghost at $z = 0$ in the disk frame, which gives $\Psi_c[c]$, we can consider $c(z) = \sum_{n \leq 1} c_n z^{1-n}$ inserted at a generic $|z| < 1$ and the corresponding $\Psi_{c(z)}[c]$, which will be built from

$$\Psi_{c_n}[c] \propto \left(\sum_{k=1}^{L} c(k) e^{2\pi i \frac{nk}{L}} \right) \prod_{m=2}^{\frac{L}{2}} \left(\sum_{k=1}^{L} c(k) e^{2\pi i \frac{mk}{L}} \right), \tag{7.11}$$

with the same identification rules (7.4) leaving the usual $\frac{L}{2}$ independent $c(k)$ variables, and resulting in an $\frac{L}{2} \times \frac{L}{2}$ matrix determinant once they are integrated over. We denote this determinant $\det B^{(n)}$, with $\det B^{(1)} = \det B$ of the preceding section.

Clearly, $\det B^{(0)} = 0$ because under (7.4), the matrix has a zero row. Numerically, we also find that $\det B^{(-n)} = 0$ for even n **[because of the symmetry of the disk under rotation by π]**. In addition, we find that in the case where all three side lengths ℓ_i are equal, $\det B^{(n)} = 0$ for all $n \neq 1 - 6k$ for some $k \in \mathbb{Z}_{\geq 0}$, corresponding to the extra $\mathbb{Z}_6$ rotational symmetry of the disk frame in this special case.

We can define

$$g(z) = 1 + \sum_{n=1}^{L/2}{}' \det B^{(-n)} z^{1+n} \tag{7.12}$$

with the $\sum'$ indicating summation over odd n. $g(z)$ is proportional to the path integral of $\Psi_{c(z)}[c]$. Considering that the boundary of the unit disc is identified under $z \to z' = q/z$, with $|q| = 1$ being three different phases for the three different segments on the boundary, it turns out (numerically tested) that

$$\frac{g(z')}{dz'} = \frac{g(z)}{dz} \quad \Leftrightarrow \quad -\frac{g(z'(z))}{z'} = \frac{g(z)}{z}, \qquad \text{for } |z| = 1. \tag{7.13}$$

Therefore

$$g(z) = (f(z))^{-1}, \tag{7.14}$$

and $g(z)$ precisely corresponds to the z-dependent conformal transformation factor of $c\widetilde{c}(z, \bar{z})$ to the correlator on the disc. The fact that there is seemingly no other z-dependence on the position of $c\widetilde{c}$ implies that there is no extra ghost number 0 bc insertions along the boundary of the disc, besides possibly at the points corresponding to the vertices themselves, in the correlator calculated by the discretized path integral of $\Psi_{c(z)}[c]$.

8 Analytic construction

8.1 Trinion vertex

The trinion vertex with the three faces of equal circumference 2π is constructed as

$$\{\Psi_1 \otimes \Psi_2 \otimes \Psi_3\}_{[1,1,1]} = \left\langle \prod_{i=1}^{3} [\Psi_i(0)]^{f_i} \right\rangle, \tag{8.1}$$

where f_i are the conformal maps

$$f_i = h^{-1} \circ R_i \circ h, \tag{8.2}$$

with

$$h(z) \equiv \frac{1 + iz}{1 - iz} \tag{8.3}$$

and

$$R_1(w) = e^{\frac{2\pi i}{3}} w^{\frac{2}{3}}, \qquad R_2(w) = w^{\frac{2}{3}}, \qquad R_1(w) = e^{-\frac{2\pi i}{3}} w^{\frac{2}{3}}. \tag{8.4}$$

This construction is analogous to Witten's OSFT cubic vertex. Note however that Ψ_i are not necessarily closed string fields, in that they are not required to obey level matching condition or being annihilated by b_0^-.

More generally, we need to consider trinion vertex of face circumferences $2\pi p_i$, $i = 1, 2, 3$, $p_i > 0$. Up to permutation, there are two qualitatively distinct cases: (1) p_1, p_2, p_3 obey

triangular inequality, in which case the ribbon graph is the theta graph, where the three edge lengths ℓ_1, ℓ_2, ℓ_3 are related by

$$2\pi p_1 = \ell_2 + \ell_3, \quad 2\pi p_2 = \ell_3 + \ell_1, \quad 2\pi p_3 = \ell_1 + \ell_2, \tag{8.5}$$

and (2) p_1, p_2, p_3 do not obey triangular inequality, in which case the ribbon is the dumbbell graph, where the three edge lengths are related by

$$2\pi p_1 = \ell_1, \quad 2\pi p_2 = \ell_2, \quad 2\pi p_3 = \ell_1 + \ell_2 + 2\ell_3. \tag{8.6}$$

We will denote the general trinion vertex by

$$\{\Psi_1 \otimes \Psi_2 \otimes \Psi_3\}_{[p_1,p_2,p_3]}. \tag{8.7}$$

To construct the explicit conformal map, we use the Strebel differential

$$\omega = \frac{P(z)}{(z-z_1)^2(z-z_2)^2(z-z_3)^2} dz^2, \tag{8.8}$$

where $P(z)$ is a quadratic polynomial in z determined by the conditions

$$\frac{P(z_1)}{z_{12}^2 z_{13}^2} = -p_1^2, \quad \frac{P(z_2)}{z_{21}^2 z_{23}^2} = -p_2^2, \quad \frac{P(z_3)}{z_{31}^2 z_{32}^2} = -p_3^2. \tag{8.9}$$

The relevant conformal maps f_i are given by

$$f_i^{-1}(z) = \exp\left(\frac{1}{p_i}\int^z \sqrt{\omega}\right), \quad z \in D_i, \tag{8.10}$$

where D_i are domains of disc topology that contain z_i, such that f_i maps the unit circle to ∂D_i. We can then write

$$\{\Psi_1 \otimes \Psi_2 \otimes \Psi_3\}_{[p_1,p_2,p_3]} = \left\langle \prod_{i=1}^{3} [\Psi_i(z_i)]^{f_i} \right\rangle. \tag{8.11}$$

8.2 General correlator associated with a ribbon graph

A general ribbon graph of genus g and n faces can be decomposed into $2g-2+n$ trinion graphs, stitched together with twists. The corresponding correlators can be determined by tracing over internal states with twists.

For instance, the $(g,n) = (0,4)$ case is given by

$$\{\Psi_1 \otimes \Psi_2 \otimes \Psi_3 \otimes \Psi_4\}^{(0,4)}_{[p_1,p_2,p_3,p_4;x,y]} = \sum_I \{\Psi_1 \otimes \Psi_2 \otimes e^{2\pi i L_0^- y}\phi_I\}_{[p_1,p_2,x]}\{\phi_I^\vee \otimes \Psi_3 \otimes \Psi_4\}_{[x,p_3,p_4]}, \tag{8.12}$$

where ϕ_I and $\phi_I^\vee$ are dual bases (with respect to BPZ inner product) of the worldsheet CFT state space. The underlying ribbon graph has 4 vertices, 6 edges, 4 faces.

For (p_1, p_2, x) and (p_3, p_4, x) obeying triangular inequalities, the 6 edge lengths ℓ_i are related by

$$\begin{aligned} 2\pi p_1 &= \ell_1 + \ell_2 + \ell_3, \\ 2\pi p_2 &= \ell_1 + \ell_4 + \ell_5, \\ 2\pi p_3 &= \ell_3 + \ell_4 + \ell_6, \\ 2\pi p_4 &= \ell_2 + \ell_5 + \ell_6, \\ 2\pi x &= \ell_2 + \ell_3 + \ell_4 + \ell_5, \\ 2\pi y &= \ell_2. \end{aligned} \tag{8.13}$$

In other cases, the underlying ribbon graph may also involve a dumbbell subgraph.

As another example, for $(g, n) = (1, 1)$, we have

$$\{\Psi\}^{(1,1)}_{[p;x,y]} = \sum_I \{\Psi \otimes e^{2\pi i L_0^- y} \phi_I \otimes \phi_I^\vee\}_{[p,x,x]}. \tag{8.14}$$

Here $p > 2x$. The underlying ribbon graph has 2 vertices, 3 edges, 1 face. It is formed out of self-plumbing of a dumbell graph, whose edge lengths $\widetilde{\ell}_1, \widetilde{\ell}_2, \widetilde{\ell}_3$ are related by

$$2\pi p = 2\widetilde{\ell}_1 + \widetilde{\ell}_2 + \widetilde{\ell}_3, \qquad 2\pi x = \widetilde{\ell}_2 = \widetilde{\ell}_3. \tag{8.15}$$

The $(g, n) = (1, 1)$ ribbon graph has three edges of length ℓ_1, ℓ_2, ℓ_3, related by

$$\ell_1 = \widetilde{\ell}_1, \qquad \ell_2 = y, \qquad \ell_3 = \widetilde{\ell}_2 - y. \tag{8.16}$$

Note that $\ell_1 + \ell_2 + \ell_3 = \widetilde{\ell}_1 + \widetilde{\ell}_2 = \pi p$.